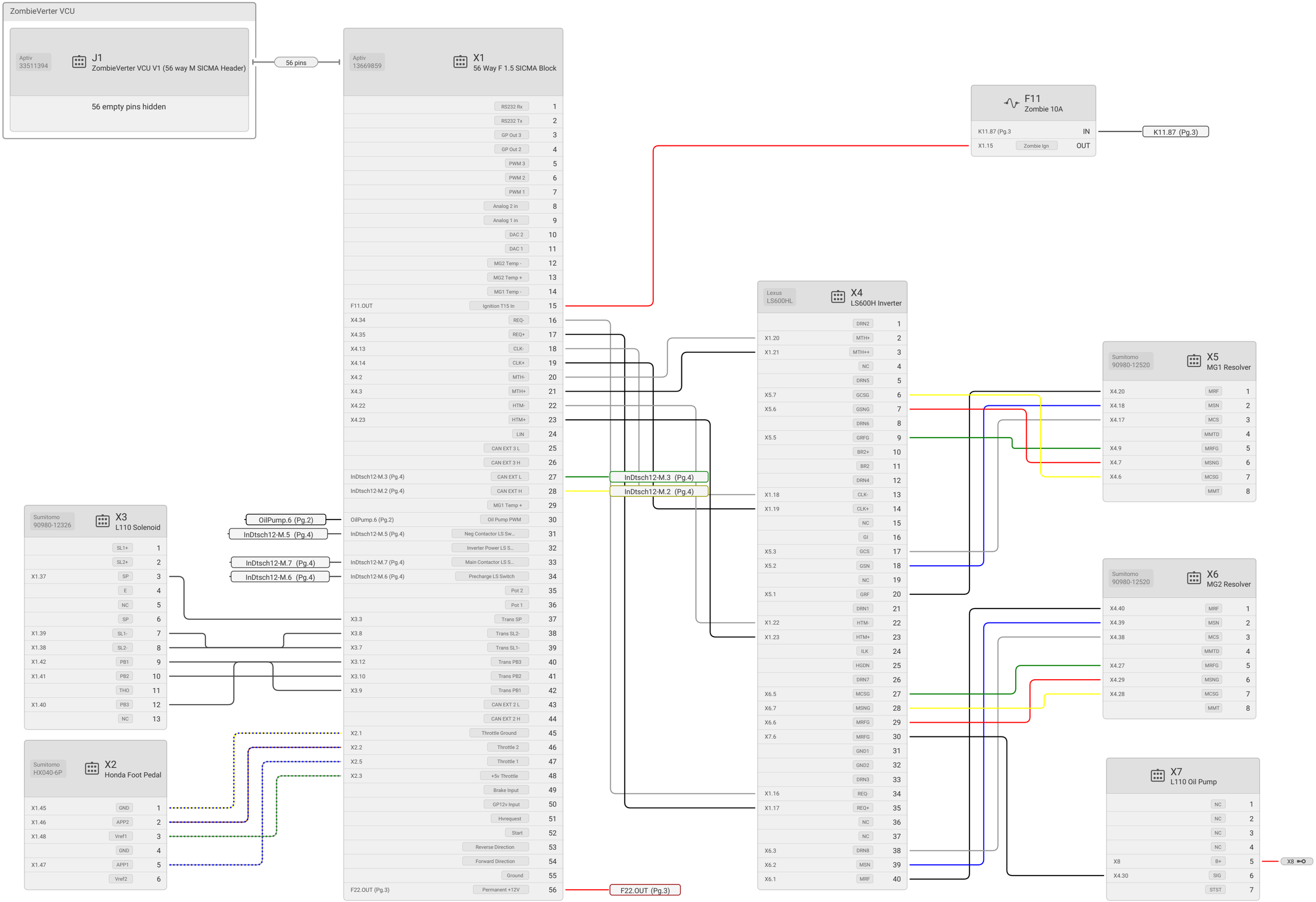
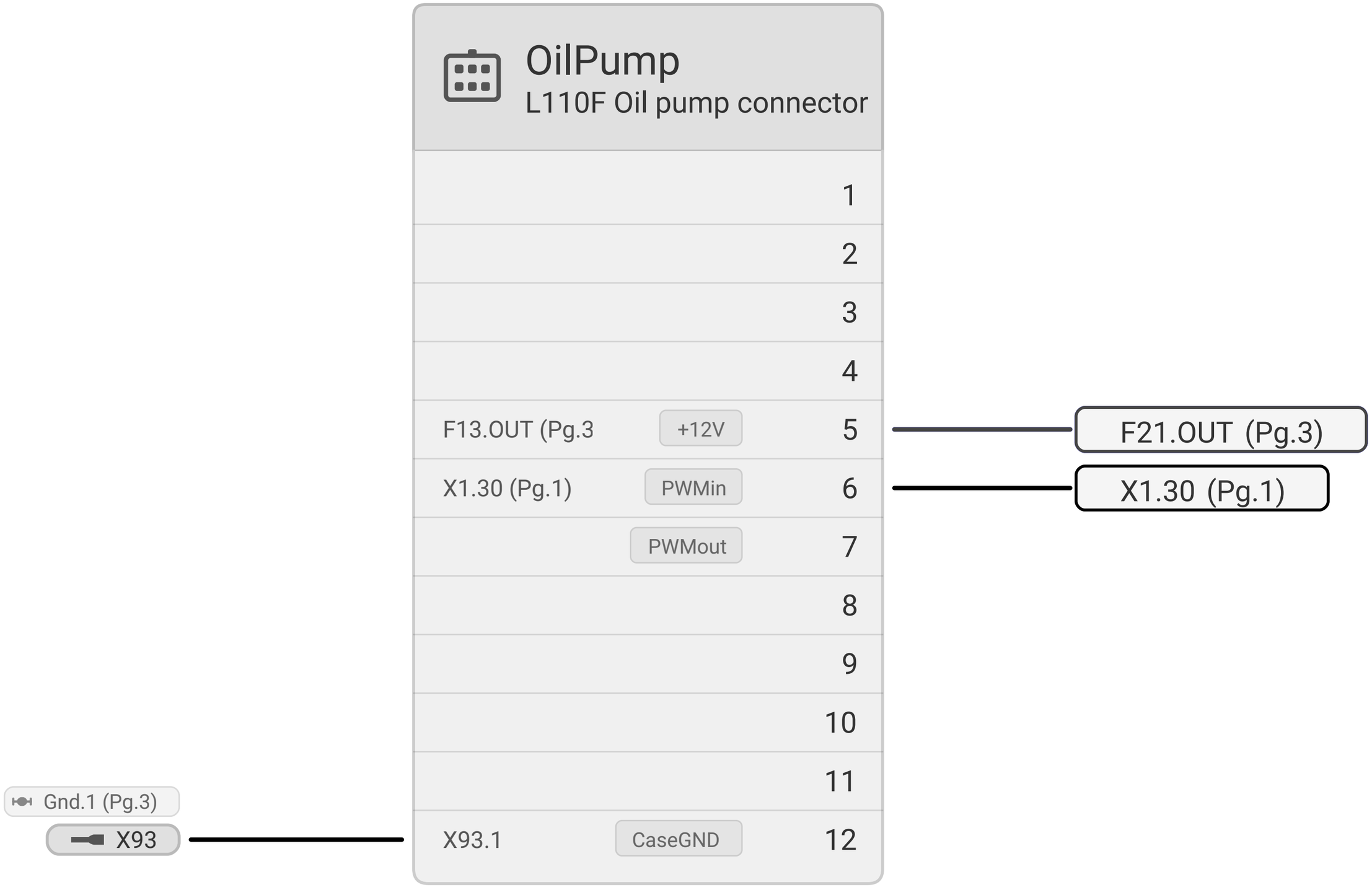
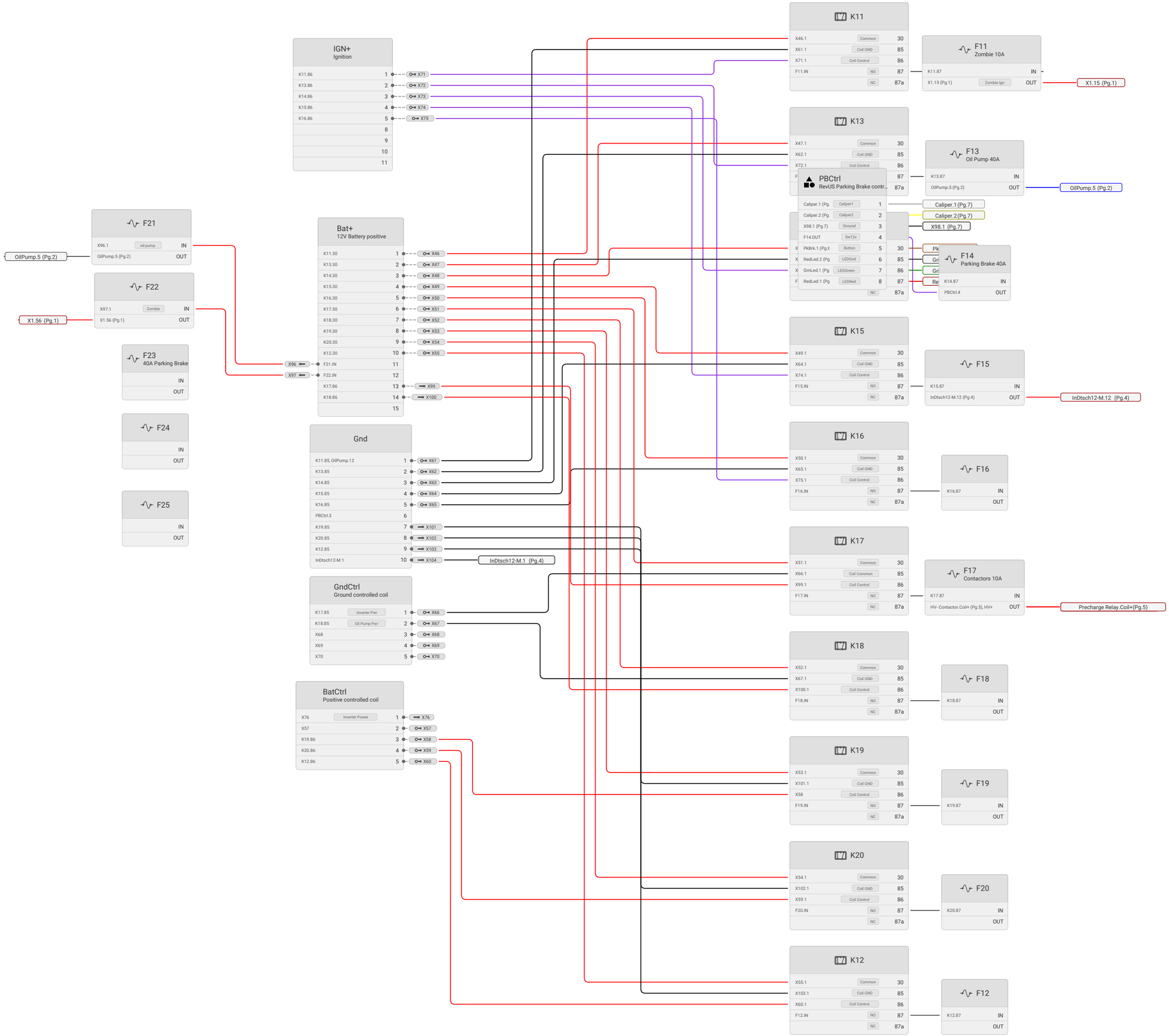
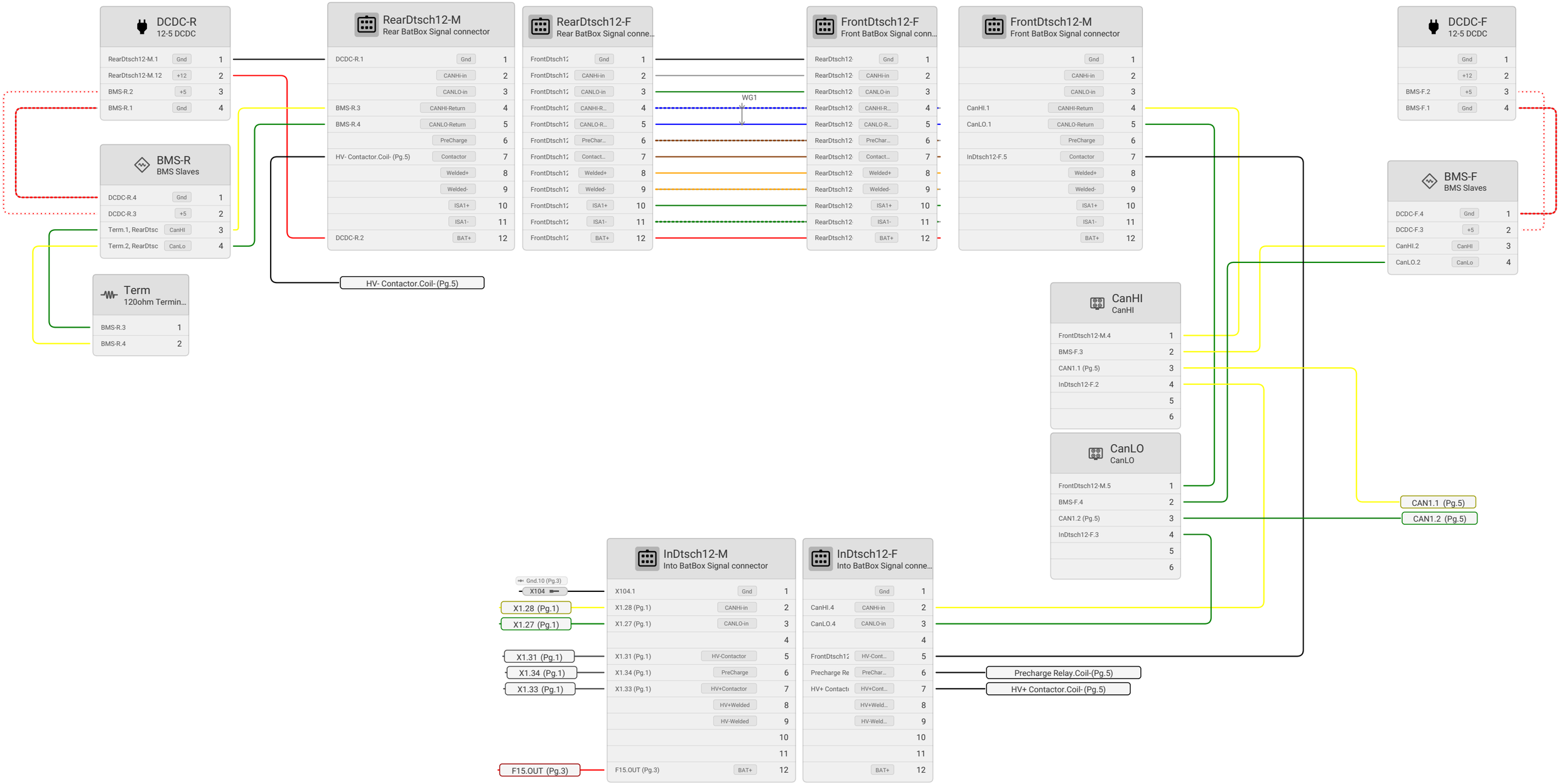
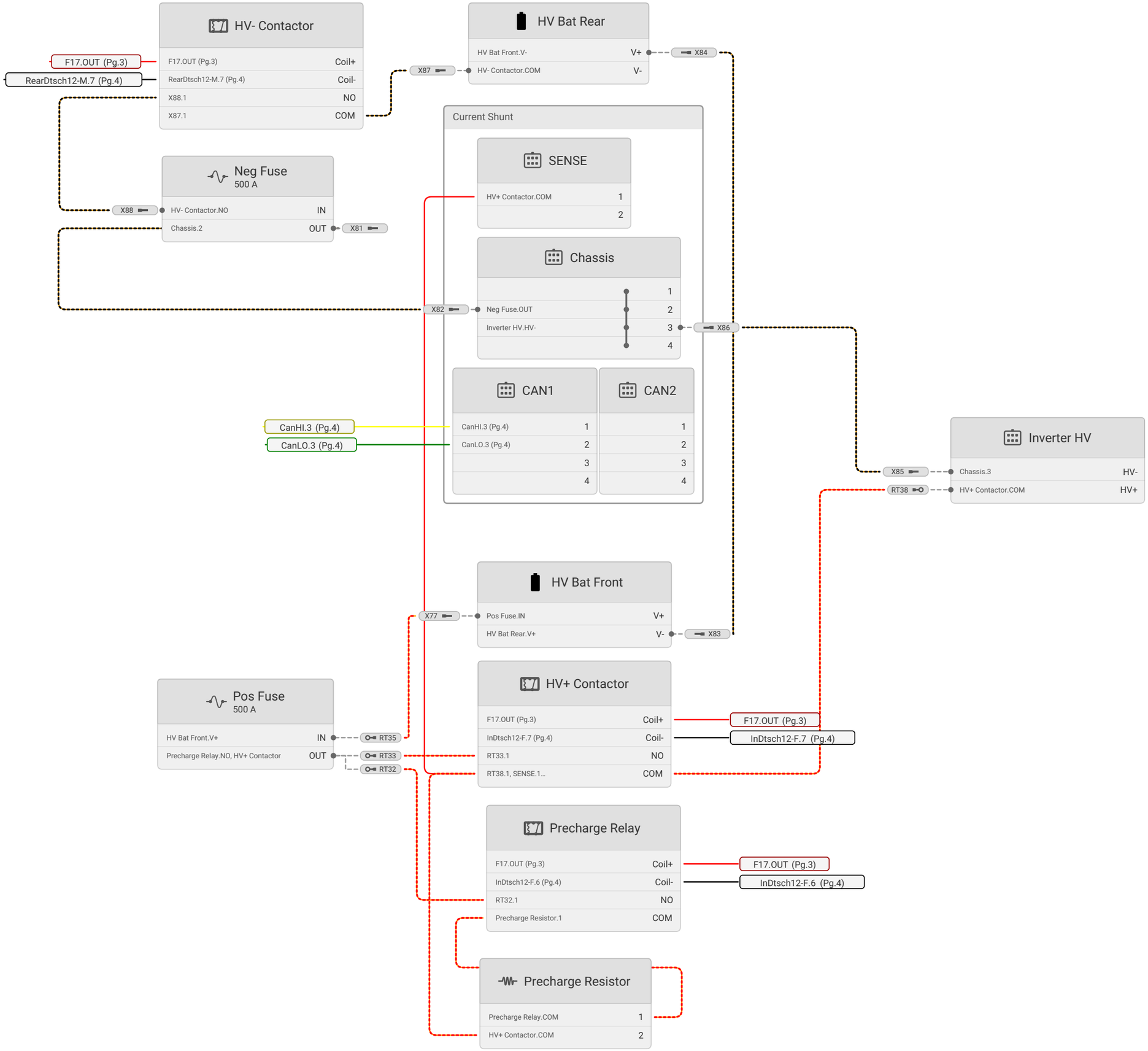
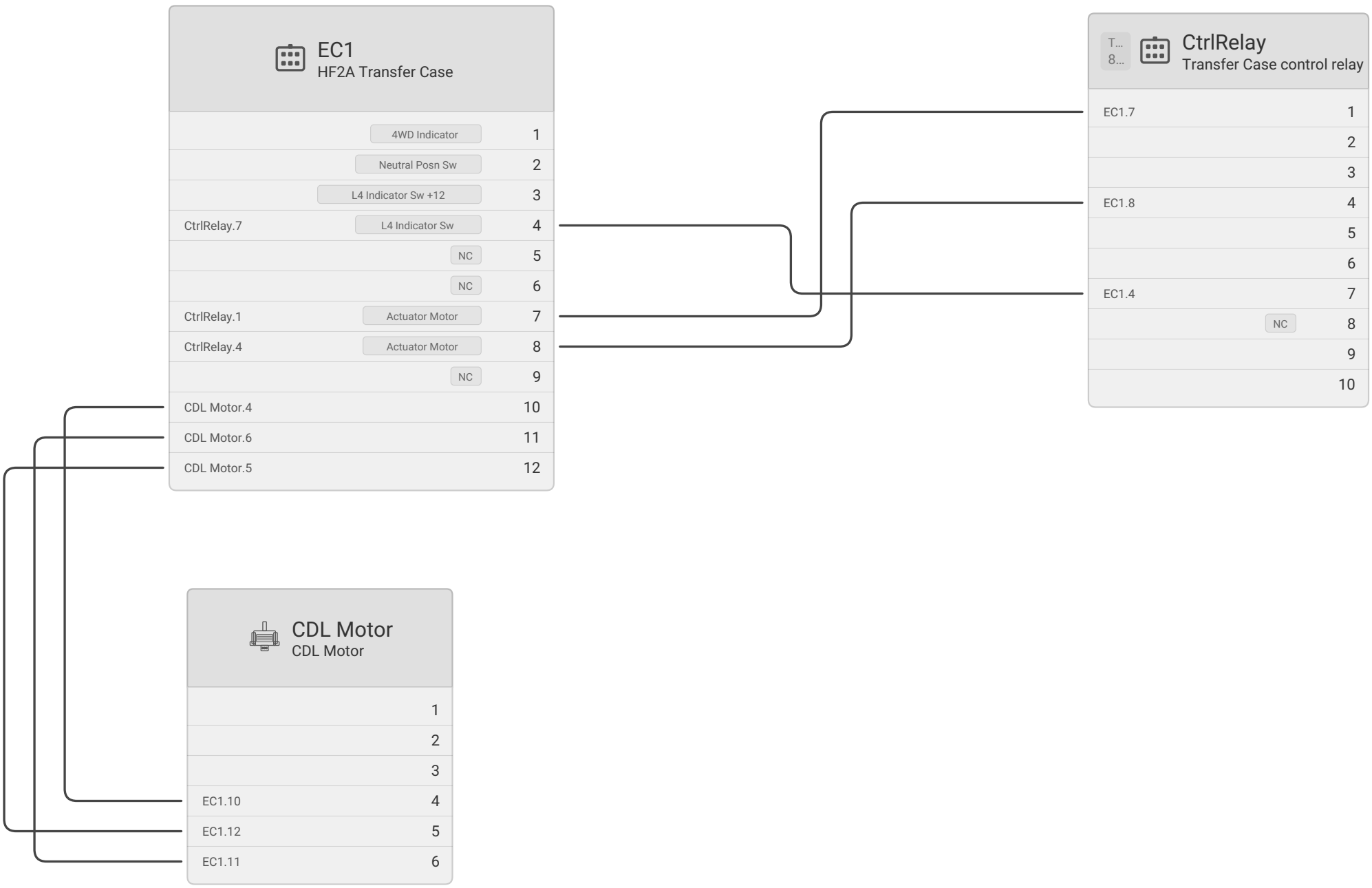

ZombieVerter VCU
Aptiv 33511394
J1
ZombieVerter VCU V1 (56 way M SICMA Header)
56 empty pins hidden
56 pins
Aptiv 13669859
X1
56 Way F 1.5 SICMA Block
1 RS232 Rx
2 RS232 Tx
3 GP Out 3
4 GP Out 2
5 PWM 3
6 PWM 2
7 PWM 1
8 Analog 2 in
9 Analog 1 in
10 DAC 2
11 DAC 1
12 MG2 Temp -
13 MG2 Temp +
14 MG1 Temp -
15 F11.OUT Ignition T15 In
16 X4.34 REQ-
17 X4.35 REQ+
18 X4.13 CLK-
19 X4.14 CLK+
20 X4.2 MTH-
21 X4.3 MTH+
22 X4.22 HTM-
23 X4.23 HTM+
24 LIN
25 CAN EXT 3 L
26 CAN EXT 3 H
27 InDtsch12-M.3 (Pg.4) CAN EXT L
28 InDtsch12-M.2 (Pg.4) CAN EXT H
29 MG1 Temp +
30 OilPump.6 (Pg.2) Oil Pump PWM
31 InDtsch12-M.5 (Pg.4) Neg Contactor LS Sw...
32 Inverter Power LS S...
33 InDtsch12-M.7 (Pg.4) Main Contactor LS S...
34 InDtsch12-M.6 (Pg.4) Precharge LS Switch
35 Pot 2
36 Pot 1
37 X3.3 Trans SP
38 X3.8 Trans SL2-
39 X3.7 Trans SL1-
40 X3.12 Trans PB3
41 X3.10 Trans PB2
42 X3.9 Trans PB1
43 CAN EXT 2 L
44 CAN EXT 2 H
45 X2.1 Throttle Ground
46 X2.2 Throttle 2
47 X2.5 Throttle 1
48 X2.3 +5v Throttle
49 Brake Input
50 GP12v Input
51 Hvrequest
52 Start
53 Reverse Direction
54 Forward Direction
55 Ground
56 F22.OUT (Pg.3) Permanent +12V
F11
Zombie 10A
K11.87 (Pg.3 IN
X1.15 Zombie Ign OUT
K11.87 (Pg.3)
Sumitomo 90980-12326
X3
L110 Solenoid
1 SL1+
2 SL2+
3 X1.37 SP
4 E
5 NC
6 SP
7 X1.39 SL1-
8 X1.38 SL2-
9 X1.42 PB1
10 X1.41 PB2
11 THO
12 X1.40 PB3
13 NC
OilPump.6 (Pg.2)
InDtsch12-M.5 (Pg.4)
InDtsch12-M.7 (Pg.4)
InDtsch12-M.6 (Pg.4)
Sumitomo HX040-6P
X2
Honda Foot Pedal
1 X1.45 GND
2 X1.46 APP2
3 X1.48 Vref1
4 GND
5 X1.47 APP1
6 Vref2
Lexus LS600HL
X4
LS600H Inverter
1 DRN2
2 X1.20 MTH+
3 X1.21 MTH++
4 NC
5 DRN5
6 X5.7 GCSG
7 X5.6 GSNG
8 DRN6
9 X5.5 GRFG
10 BR2+
11 BR2
12 DRN4
13 X1.18 CLK-
14 X1.19 CLK+
15 NC
16 GI
17 X5.3 GCS
18 X5.2 GSN
19 NC
20 X5.1 GRF
21 DRN1
22 X1.22 HTM-
23 X1.23 HTM+
24 ILK
25 HGDN
26 DRN7
27 X6.5 MCSG
28 X6.7 MSNG
29 X6.6 MRFG
30 X7.6 MRFG
31 GND1
32 GND2
33 DRN3
34 X1.16 REQ-
35 X1.17 REQ+
36 NC
37 NC
38 X6.3 DRN8
39 X6.2 MSN
40 X6.1 MRF
InDtsch12-M.3 (Pg.4)
InDtsch12-M.2 (Pg.4)
F22.OUT (Pg.3)
Sumitomo 90980-12520
X5
MG1 Resolver
1 X4.20 MRF
2 X4.18 MSN
3 X4.17 MCS
4 MMTD
5 X4.9 MRFG
6 X4.7 MSNG
7 X4.6 MCSG
8 MMT
Sumitomo 90980-12520
X6
MG2 Resolver
1 X4.40 MRF
2 X4.39 MSN
3 X4.38 MCS
4 MMTD
5 X4.27 MRFG
6 X4.29 MSNG
7 X4.28 MCSG
8 MMT
X7
L110 Oil Pump
1 NC
2 NC
3 NC
4 NC
5 X8 B+
6 X4.30 SIG
7 STST
X8

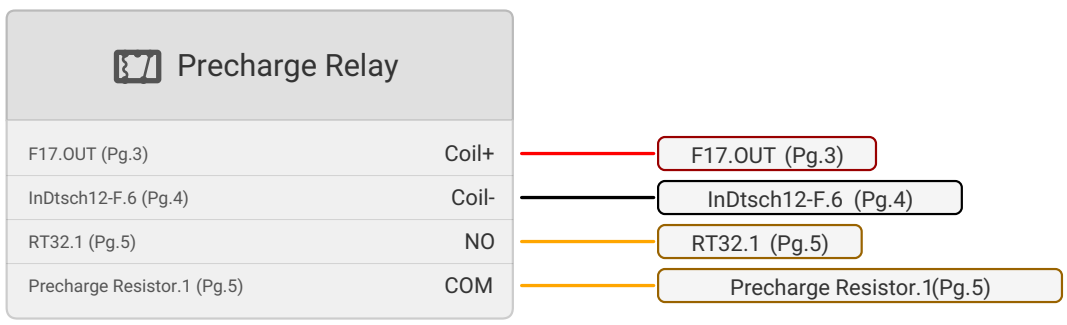

OilPump
L110F Oil pump connector
1
2
3
4
F13.OUT (Pg.3
+12V
5
F21.OUT (Pg.3)
X1.30 (Pg.1)
PWMin
6
X1.30 (Pg.1)
PWMout
7
8
9
10
11
Gnd.1 (Pg.3)
X93
X93.1
CaseGND
12

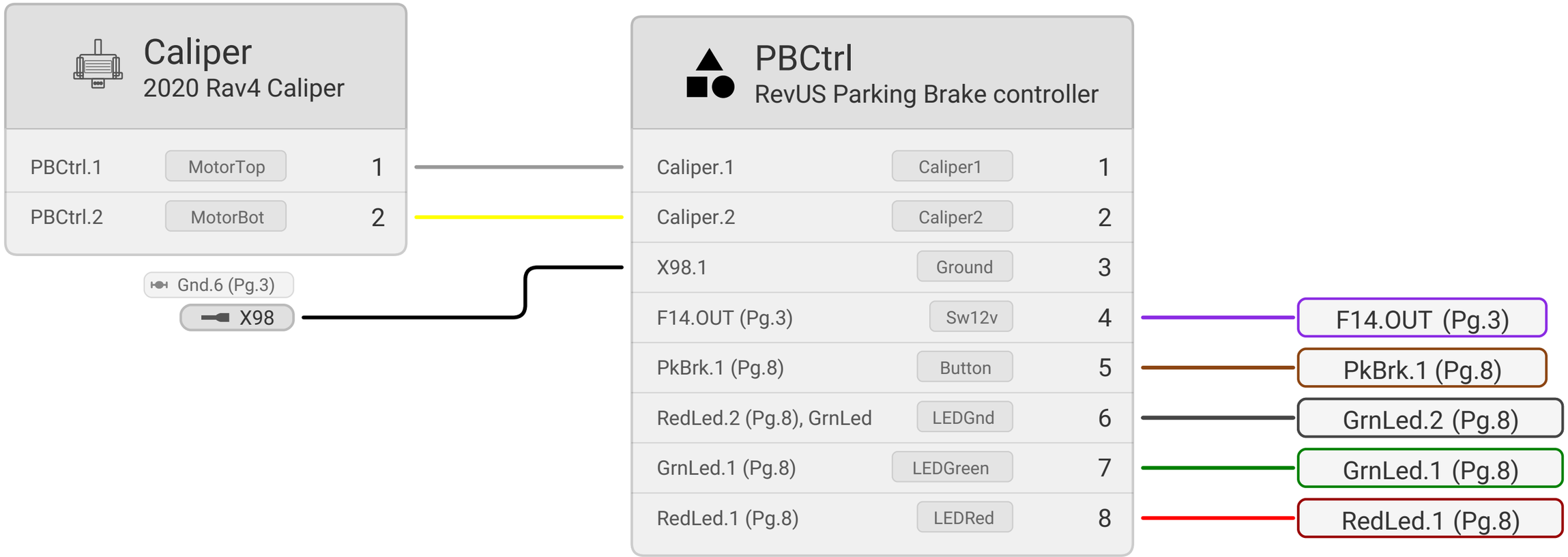

IGN+
Ignition
K11.86 1 X71
K13.86 2 X72
K14.86 3 X73
K15.86 4 X74
K16.86 5 X75
8
9
10
11
Bat+
12V Battery positive
K11.30 1 X46
K13.30 2 X47
K14.30 3 X48
K15.30 4 X49
K16.30 5 X50
K17.30 6 X51
K18.30 7 X52
K19.30 8 X53
K20.30 9 X54
K12.30 10 X55
F21.IN 11 X96
F22.IN 12 X97
K17.86 13 X99
K18.86 14 X100
15
Gnd
K11.85, OilPump.12 1 X61
K13.85 2 X62
K14.85 3 X63
K15.85 4 X64
K16.85 5 X65
PBCtrl.3 6
K19.85 7 X101
K20.85 8 X102
K12.85 9 X103
InDtsch12-M.1 10 X104
InDtsch12-M.1 (Pg.4)
GndCtrl
Ground controlled coil
K17.85 Inverter Pwr 1 X66
K18.85 Oil Pump Pwr 2 X67
X68 3 X68
X69 4 X69
X70 5 X70
BatCtrl
Positive controlled coil
X76 Inverter Power 1 X76
X57 2 X57
K19.86 3 X58
K20.86 4 X59
K12.86 5 X60
F21
X96.1 oil pump IN
OilPump.5 (Pg.2) OUT
OilPump.5 (Pg.2)
F22
X97.1 Zombie IN
X1.56 (Pg.1) OUT
X1.56 (Pg.1)
F23
40A Parking Brake
IN
OUT
F24
IN
OUT
F25
IN
OUT
K11
X46.1 Common 30
X61.1 Coil GND 85
X71.1 Coil Control 86
F11.IN NO 87
NC 87a
F11
Zombie 10A
K11.87 IN
X1.15 (Pg.1) Zombie Ign OUT
X1.15 (Pg.1)
K13
X47.1 Common 30
X62.1 Coil GND 85
X72.1 Coil Control 86
87
87a
F13
Oil Pump 40A
K13.87 IN
OilPump.5 (Pg.2) OUT
OilPump.5 (Pg.2)
PBCtrl
RevUS Parking Brake contr...
Caliper.1 (Pg Caliper1 1
Caliper.2 (Pg Caliper2 2
X98.1 (Pg.7) Ground 3
F14.OUT Sw12v 4
PkBrk.1 (Pg.8 Button 5 30
RedLed.2 (Pg LEDGnd 6 85
GrnLed.1 (Pg LEDGreen 7 86
RedLed.1 (Pg LEDRed 8 87
NC 87a
Caliper.1 (Pg.7)
Caliper.2 (Pg.7)
X98.1 (Pg.7)
F14
Parking Brake 40A
K14.87 IN
PBCtrl.4 OUT
K15
X49.1 Common 30
X64.1 Coil GND 85
X74.1 Coil Control 86
F15.IN NO 87
NC 87a
F15
K15.87 IN
InDtsch12-M.12 (Pg.4) OUT
InDtsch12-M.12 (Pg.4)
K16
X50.1 Common 30
X65.1 Coil GND 85
X75.1 Coil Control 86
F16.IN NO 87
NC 87a
F16
K16.87 IN
OUT
K17
X51.1 Common 30
X66.1 Coil Common 85
X99.1 Coil Control 86
F17.IN NO 87
NC 87a
F17
Contactors 10A
K17.87 IN
HV- Contactor.Coil+ (Pg.5), HV+ OUT
Precharge Relay.Coil+(Pg.5)
K18
X52.1 Common 30
X67.1 Coil GND 85
X100.1 Coil Control 86
F18.IN NO 87
NC 87a
F18
K18.87 IN
OUT
K19
X53.1 Common 30
X101.1 Coil GND 85
X58 Coil Control 86
F19.IN NO 87
NC 87a
F19
K19.87 IN
OUT
K20
X54.1 Common 30
X102.1 Coil GND 85
X59.1 Coil Control 86
F20.IN NO 87
NC 87a
F20
K20.87 IN
OUT
K12
X55.1 Common 30
X103.1 Coil GND 85
X60.1 Coil Control 86
F12.IN NO 87
NC 87a
F12
K12.87 IN
OUT

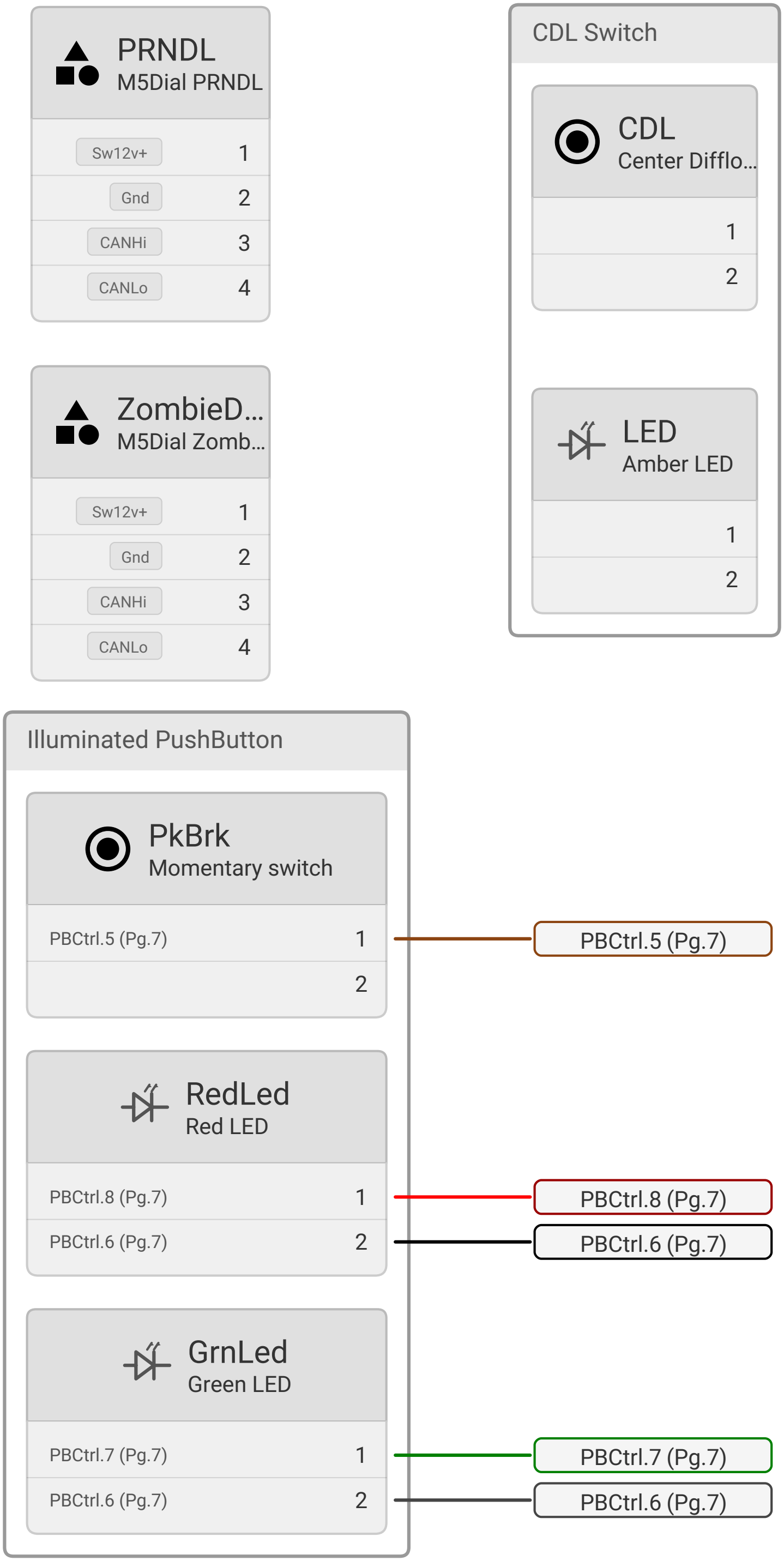

DCDC-R
12-5 DCDC
RearDtsch12-M.1 Gnd 1
RearDtsch12-M.12 +12 2
BMS-R.2 +5 3
BMS-R.1 Gnd 4
BMS-R
BMS Slaves
DCDC-R.4 Gnd 1
DCDC-R.3 +5 2
Term.1, RearDtsc CanHI 3
Term.2, RearDtsc CanLo 4
Term
120ohm Termin...
BMS-R.3 1
BMS-R.4 2
RearDtsch12-M
Rear BatBox Signal connector
DCDC-R.1 Gnd 1
CANHi-in 2
CANLO-in 3
BMS-R.3 CANHI-Return 4
BMS-R.4 CANLO-Return 5
PreCharge 6
HV- Contactor.Coil- (Pg.5) Contactor 7
Welded+ 8
Welded- 9
ISA1+ 10
ISA1- 11
DCDC-R.2 BAT+ 12
HV- Contactor.Coil- (Pg.5)
RearDtsch12-F
Rear BatBox Signal conne...
FrontDtsch1 Gnd 1
FrontDtsch1 CANHi-in 2
FrontDtsch1 CANLO-in 3
FrontDtsch1 CANHI-R... 4
FrontDtsch1 CANLO-R... 5
FrontDtsch1 PreChar... 6
FrontDtsch1 Contact... 7
FrontDtsch1 Welded+ 8
FrontDtsch1 Welded- 9
FrontDtsch1 ISA1+ 10
FrontDtsch1 ISA1- 11
FrontDtsch1 BAT+ 12
WG1
FrontDtsch12-F
Front BatBox Signal conn...
RearDtsch12 Gnd 1
RearDtsch12 CANHi-in 2
RearDtsch12 CANLO-in 3
RearDtsch12 CANHI-R... 4
RearDtsch12 CANLO-R... 5
RearDtsch12 PreChar... 6
RearDtsch12 Contact... 7
RearDtsch12 Welded+ 8
RearDtsch12 Welded- 9
RearDtsch12 ISA1+ 10
RearDtsch12 ISA1- 11
RearDtsch12 BAT+ 12
FrontDtsch12-M
Front BatBox Signal connector
Gnd 1
CANHi-in 2
CANLO-in 3
CanHI.1 CANHI-Return 4
CanLO.1 CANLO-Return 5
PreCharge 6
InDtsch12-F.5 Contactor 7
Welded+ 8
Welded- 9
ISA1+ 10
ISA1- 11
BAT+ 12
DCDC-F
12-5 DCDC
Gnd 1
+12 2
BMS-F.2 +5 3
BMS-F.1 Gnd 4
BMS-F
BMS Slaves
DCDC-F.4 Gnd 1
DCDC-F.3 +5 2
CanHI.2 CanHI 3
CanLO.2 CanLo 4
CanHI
CanHI
FrontDtsch12-M.4 1
BMS-F.3 2
CAN1.1 (Pg.5) 3
InDtsch12-F.2 4
5
6
CanLO
CanLO
FrontDtsch12-M.5 1
BMS-F.4 2
CAN1.2 (Pg.5) 3
InDtsch12-F.3 4
5
6
CAN1.1 (Pg.5)
CAN1.2 (Pg.5)
Gnd.10 (Pg.3)
X104
X1.28 (Pg.1)
X1.27 (Pg.1)
X1.31 (Pg.1)
X1.34 (Pg.1)
X1.33 (Pg.1)
F15.OUT (Pg.3)
InDtsch12-M
Into BatBox Signal connector
X104.1 Gnd 1
X1.28 (Pg.1) CANHi-in 2
X1.27 (Pg.1) CANLO-in 3
4
X1.31 (Pg.1) HV-Contactor 5
X1.34 (Pg.1) PreCharge 6
X1.33 (Pg.1) HV+Contactor 7
HV+Welded 8
HV-Welded 9
10
11
F15.OUT (Pg.3) BAT+ 12
InDtsch12-F
Into BatBox Signal conne...
Gnd 1
CanHI.4 CANHi-in 2
CanLO.4 CANLO-in 3
4
FrontDtsch12 HV-Cont... 5
Precharge Re PreChar... 6
HV+ Contacto HV+Cont... 7
HV+Weld... 8
HV-Weld... 9
10
11
BAT+ 12
Precharge Relay.Coil-(Pg.5)
HV+ Contactor.Coil- (Pg.5)


HV- Contactor
F17.OUT (Pg.3)
RearDtsch12-M.7 (Pg.4)
F17.OUT (Pg.3) Coil+
RearDtsch12-M.7 (Pg.4) Coil-
X88.1 NO
X87.1 COM
HV Bat Rear
HV Bat Front.V- V+
HV- Contactor.COM V-
X87
X84
Neg Fuse
500 A
X88
HV- Contactor.NO IN
Chassis.2 OUT
X81
Current Shunt
SENSE
HV+ Contactor.COM 1
2
Chassis
X82
Neg Fuse.OUT 1
2
Inverter HV.HV- 3
4
X86
CAN1
CAN2
CanHI.3 (Pg.4)
CanLO.3 (Pg.4)
CanHI.3 (Pg.4) 1
CanLO.3 (Pg.4) 2
3
4
Inverter HV
X85
Chassis.3 HV-
RT38
HV+ Contactor.COM HV+
HV Bat Front
X77
Pos Fuse.IN V+
HV Bat Rear.V+ V-
X83
Pos Fuse
500 A
HV Bat Front.V+ IN
Precharge Relay.NO, HV+ Contactor OUT
RT35
RT33
RT32
HV+ Contactor
F17.OUT (Pg.3) Coil+
InDtsch12-F.7 (Pg.4) Coil-
RT33.1 NO
RT38.1, SENSE.1... COM
F17.OUT (Pg.3)
InDtsch12-F.7 (Pg.4)
Precharge Relay
F17.OUT (Pg.3) Coil+
InDtsch12-F.6 (Pg.4) Coil-
RT32.1 NO
Precharge Resistor.1 COM
F17.OUT (Pg.3)
InDtsch12-F.6 (Pg.4)
Precharge Resistor
Precharge Relay.COM 1
HV+ Contactor.COM 2


EC1
HF2A Transfer Case
4WD Indicator 1
Neutral Posn Sw 2
L4 Indicator Sw +12 3
CtrlRelay.7 L4 Indicator Sw 4
NC 5
NC 6
CtrlRelay.1 Actuator Motor 7
CtrlRelay.4 Actuator Motor 8
NC 9
CDL Motor.4 10
CDL Motor.6 11
CDL Motor.5 12
T... 8...
CtrlRelay
Transfer Case control relay
EC1.7 1
2
3
EC1.8 4
5
6
EC1.4 7
NC 8
9
10
CDL Motor
CDL Motor
1
2
3
EC1.10 4
EC1.12 5
EC1.11 6


Precharge Relay
F17.OUT (Pg.3) Coil+
InDtsch12-F.6 (Pg.4) Coil-
RT32.1 (Pg.5) NO
Precharge Resistor.1 (Pg.5) COM
F17.OUT (Pg.3)
InDtsch12-F.6 (Pg.4)
RT32.1 (Pg.5)
Precharge Resistor.1(Pg.5)


Caliper
2020 Rav4 Caliper
PBCtrl.1
MotorTop
1
PBCtrl.2
MotorBot
2
Gnd.6 (Pg.3)
X98
PBCtrl
RevUS Parking Brake controller
Caliper.1
Caliper1
1
Caliper.2
Caliper2
2
X98.1
Ground
3
F14.OUT (Pg.3)
Sw12v
4
PkBrk.1 (Pg.8)
Button
5
RedLed.2 (Pg.8), GrnLed
LEDGnd
6
GrnLed.1 (Pg.8)
LEDGreen
7
RedLed.1 (Pg.8)
LEDRed
8
F14.OUT (Pg.3)
PkBrk.1 (Pg.8)
GrnLed.2 (Pg.8)
GrnLed.1 (Pg.8)
RedLed.1 (Pg.8)


PRNDL
M5Dial PRNDL
Sw12v+ 1
Gnd 2
CANHi 3
CANLo 4
ZombieD...
M5Dial Zomb...
Sw12v+ 1
Gnd 2
CANHi 3
CANLo 4
CDL Switch
CDL
Center Difflo...
1
2
LED
Amber LED
1
2
Illuminated PushButton
PkBrk
Momentary switch
PBCtrl.5 (Pg.7) 1
2
PBCtrl.5 (Pg.7)
RedLed
Red LED
PBCtrl.8 (Pg.7) 1
PBCtrl.6 (Pg.7) 2
PBCtrl.8 (Pg.7)
PBCtrl.6 (Pg.7)
GrnLed
Green LED
PBCtrl.7 (Pg.7) 1
PBCtrl.6 (Pg.7) 2
PBCtrl.7 (Pg.7)
PBCtrl.6 (Pg.7)